

GATE QUESTION ECE 2010 Q56

Question :

25 persons are in a room. 15 of them play hockey, 17 of them play football and 10 of them play both hockey and football. Then the number of persons playing neither hockey nor football is

1. 2
2. 17
3. 13
4. 3

Solution:

This problem can be solved using the principle of inclusion-exclusion in set theory.

Let N be the total number of persons in the room. $N = 25$

Let H be the set of persons who play hockey. $|H| = 15$

Let F be the set of persons who play football. $|F| = 17$

Let $H \cap F$ be the set of persons who play both hockey and football. $|H \cap F| = 10$

We want to find the number of persons who play neither hockey nor football. This can be found by subtracting the number of persons who play at least one of the sports from the total number of persons.

First, calculate the number of persons who play at least one sport (hockey OR football), which is represented by the union of the two sets, $|H \cup F|$. According to the principle of inclusion-exclusion:

$$|H \cup F| = |H| + |F| - |H \cap F|$$

Substitute the given values into the formula:

$$|H \cup F| = 15 + 17 - 10$$

$$|H \cup F| = 22$$

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So, 22 persons play at least one of the two sports (hockey or football).

Now, to find the number of persons who play neither hockey nor football, we subtract the number of persons playing at least one sport from the total number of persons:

$$\text{Number of persons playing neither} = N - |H \cup F|$$

$$\text{Number of persons playing neither} = 25 - 22$$

$$\text{Number of persons playing neither} = 3$$

Final Answer:

The number of persons playing neither hockey nor football is 3. This corresponds to option (D).